HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 419: Communication Engineering

Batch : 24th
Date: 06/09/2026

Final Examination

Semester: Spring 2026
Time: 2 Hours
Total Marks : 40

Answer any **five** questions: $5 \times 8 = 40$

		CLOs	Marks
1.	(a) Explain the differences between a fixed WiMAX and a mobile WiMAX.	CLO 3	4
	(b) Compare a piconet and a scatternet in the Bluetooth architecture.	CLO 3	4
2.	(a) Write the relationship between D-AMPS and AMPS.	CLO 1	4
	(b) Compare an uplink with a downlink in satellite communication.	CLO 3	4
3.	(a) List the objectives defined by the 4G working group.	CLO 3	4
	(b) Distinguish between a hard handoff and a soft handoff.	CLO 3	4
4.	(a) Analyze why is SNR in a wireless LAN normally lower than SNR in a wired LAN?	CLO 1	4
	(b) Identify some reasons that CSMA/CD cannot be used in a wireless LAN.	CLO 3	4
5.	(a) Describe the SS7 service and its relation to the telephone network.	CLO 1	4
	(b) How data transfer is achieved using CATV channels? Explain.	CLO 3	4
6.	(a) In ATM, what are the relationship among TPs, VPs and VCs?	CLO 3	4
	(b) What type of topology is used when customers in an area use DSL modems for data transfer purposes? Explain.	CLO 3	4
7.	(a) Draw a table for SONET/SDH rates.	CLO 5	4
	(b) Briefly describe the functions of each SONET layer.	CLO 5	4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 411: Artificial Intelligence

Batch: 24th | Semester: Spring 2026
Date: 01/09/2026 | **Final Examination** | Time: 2 hours | Total Marks: 40

Answer any **five** questions from the following. 5x8= 40

1. (a) Define a regression model. Briefly state its main types, applications, and any two advantages and disadvantages. CLO 1 & CLO 2 3
   (b) Given that: CLO 4 3

Attendance (%)	IQ	Marks
60	90	55
65	95	60
70	100	65
75	105	70
80	110	78
85	115	85

   Considering IQ is not significantly related with Marks, what will be the Regression Model and model params?
   (c) Predict the Marks for Attendance 72% and IQ of 100. CLO 3 2

2. (a) Define a Nearest Neighbor (NN) model. Briefly state its main types, applications, and any two advantages and disadvantages. CLO 1 3
   (b) Given that: CLO 3 5

Attendance (%)	IQ	Marks
60	90	55
65	95	60
70	100	65
80	110	78
85	115	85

   For Attendance = 75% and IQ = 105. Predict the Mark.

3. (a) Define a Support Vector Machine (SVM). Briefly state its main types, applications, and any two advantages and disadvantages. CLO 1 & COL2 3
   (b) Given that: CLO 3 5

Attendance (%)	IQ	Marks	Class
60	90	55	Bad
65	95	60	Bad
70	100	65	Bad
80	110	78	Good
85	115	85	Good

   Compute the support vectors.

4. (a) Define a Decision Tree (DT). Briefly state its main types, applications, and any two advantages and disadvantages. COL1 & COL2 3

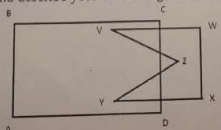
HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE413: Computer Graphics

Batch: 24th | Semester: Spring 2026
Date: 3/9/2026 | **Final Examination** | Time: 2 Hours | Total Marks: 40

Answer any five questions of the following. $5 \times 8 = 40$

			CLO	
1.	a)	Evaluate the claim that reflection is a special case of scaling, providing mathematical proof to support your conclusion.	CLO4	3
	b)	Derive an equation for reflection along any line.	CLO2	5
2.	a)	Write down the midpoint circle drawing algorithm.	CLO2	4
	b)	Using midpoint circle drawing algorithm find all the points in fourth quadrant of the circle of radius 10 and centered at origin.	CLO4	4
3.	a)	What is clipping? Write down the point clipping algorithm.	CLO1	2
	b)	Given a line segment with endpoints P1(30, 60) and P2(120, 100), and a clipping window with boundaries: $x_{min} = 40$, $y_{min} = 40$, $x_{max} = 100$, $y_{max} = 90$. Using the Cohen-Sutherland line clipping algorithm, compute the region codes for both endpoints and determine the visible portion of the line. Show all steps.	CLO2	6
4.		Define projection? Classify projection with proper example.	CLO3	8
5.		A cube has vertices A(0,0,0), B(1,0,0), C(1,1,0), D(0,1,0) on its base. Apply the following sequence of transformations in order: i. Scaling with Sx = 2, Sy = 2, Sz = 2 about the origin ii. Rotation by 90° about the Z-axis iii. Translation by Tx = 3, Ty = 2, Tz = 1 Find the final coordinates of all four vertices after the composite transformation.	CLO2	8
6.		A polygon with vertices (10,10), (60,10), (60,60), (10,60) is to be clipped against a clipping window defined by (20,20) and (50,50). Apply the Sutherland-Hodgman polygon clipping algorithm step-by-step (clip against each of the four edges: left, right, top, bottom) and show the resulting clipped polygon vertices.	CLO2	8
7.		Which polygon clipping algorithm would you choose for the given scenario? Justify your answer and describe your chosen algorithm in details.	CLO4	8



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE415: Digital Signal Processing
Final Examination
Spring 2026

Date: 08/09/2026

Time: 2 Hours

Answer any **five** questions of the followings

Total Marks: 40

No.	Question/ Statement	CLOs	Marks
1.	a) Define the Z-transform.	CLO2	2
	b) Prove the linearity property of the Z-transform.	CLO2	6
2.	a) Find the Z-transform: $\sin(3n+5)$	CLO2	4
	b) Find the Z-transform: $\cos(n\pi/2 + \pi/4)$	CLO2	4
3.	a) Define Linear Time-Invariant(LTI) System.	CLO1	2
	b) Design an FIR filter using a Hamming window for the given frequency response: $H_d(\omega) = e^{(-j3\omega)}, \quad -\pi/4 < \omega < \pi/4$	CLO3	6
4.	a) Define Impulse Response of an LTI System.	CLO1	2
	b) Design a linear-phase FIR filter of length 7 with a cut-off frequency of (1) rad/sec using a rectangular window.	CLO3	6
5.	a) Describe the basic features of IIR (Infinite Impulse Response) filters.	CLO3	4
	b) Discuss the advantages and disadvantages of IIR filters.	CLO3	4
6.	a) Differentiate between Linear Time-Invariant (LTI) systems and Linear Shift-Invariant (LSI) systems.	CLO1	4
	b) Define Convolution and write its mathematical expression.	CLO2	4
7.	a) Describe the properties of a Linear Time-Invariant (LTI) system.	CLO1	4
	b) Define a window function and explain the different types of window functions used in FIR filter design with their mathematical expressions.	CLO3	4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 417: Compiler Design

Batch: 24th
Date: 10/09/2026

Final Examination

Time: 2 hours
Total Marks: 40

Answer any five questions. **Marks: 5 * 8 = 40**

1.	(a)	Define Context-free grammar and Recursive grammar.	CLO3	2
	(b)	Write the differences between SDD and SDT.	CLO3	3
	(c)	State the summary of a recursive descent parsing in a compiler.	CLO3	3
2.	(a)	Write down about Ambiguous grammar and Unambiguous grammar.	CLO3	2
	(b)	Write a C program to sum two numbers. Then generate a symbol table for the program.	CLO3	3
	(c)	S →AA A → aA/b Find out whether the above grammar is accepted by LL(1) parser or not.	CLO3	3
3.	(a)	Write about an annotated tree.	CLO3	2
	(b)	Make the table for Quadruples for the expression : k = c* - b + c * - c +a	CLO2	3
	(c)	S →AAB A → aA|b B → jkl| tuv Find out whether the above grammar is accepted by SLR(1) parser or not.	CLO3	3
4.	(a)	State the differences between an annotated parse tree and a syntax tree.	CLO3	2
	(b)	Draw an annotated tree for : (9+8*(7+6))*4n .	CLO4	6
5.	(a)	State the rules for Quadruples.	CLO3	3
	(b)	S → S # A /A A → A & B / B B → id Given String : 10 & 4 # 6 . Draw annotate tree from the given grammar and show the output for the given string.	CLO4	5
6.	(a)	E → E + T E→ T T → T * F T →F F → id Draw the dependency graph for: 10 * 4 + 3 * 7 + 8.	CLO3	3
	(b)	S →AAB	CLO3	5

HAMDARD UNIVERSITY BANGLADESH

A → aA|C
B → jkl| tuv
C → xyz

Find out whether the above grammar is accepted by a CLR(1) parser or not. Write down why a CLR (1) parser is better than an SLR(1) parser or an LL(1) parser.

7. CLO3 8

S →AB
A → Aab|C
B → prs
C → xyz

Find out whether the above grammar is accepted by LALR(1) parser or not, and also write down why a CLR(1) parser is not an LALR(1) parser?